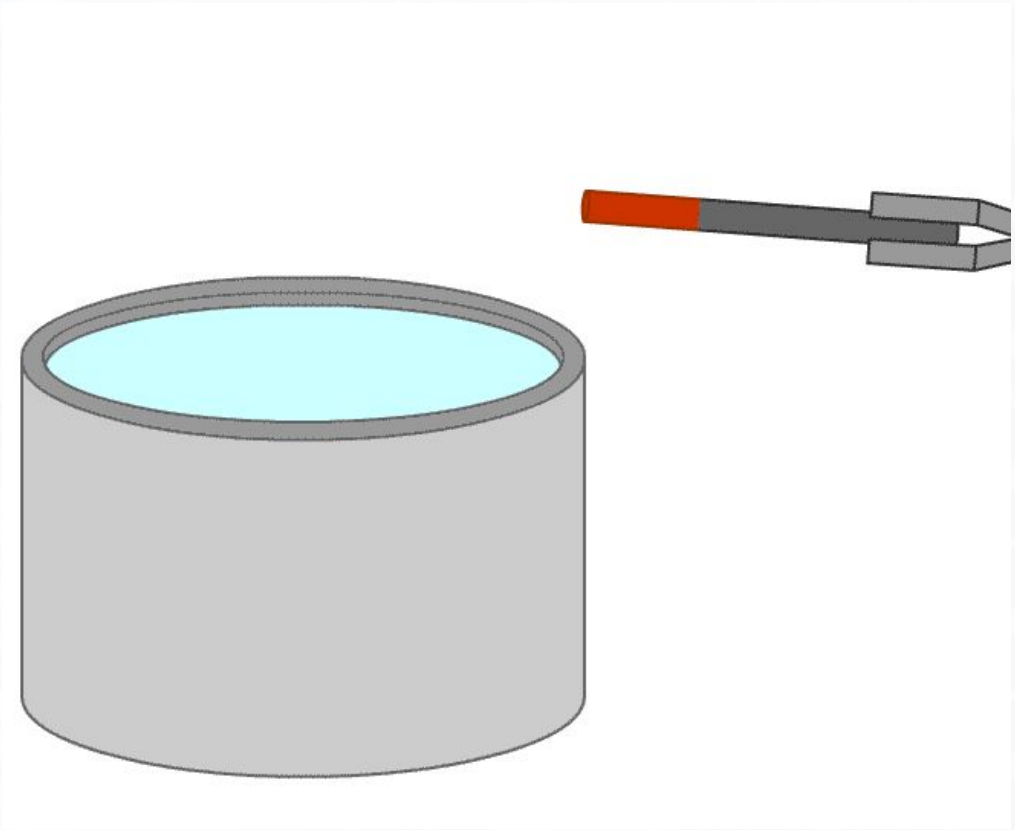
Mr Dimelow

3.1 Engineering materials

3.1.1.1 Metals and alloys

Subject content	Additional information	Mathematical understanding
Ferrous metals and alloys: • cast iron • low and high carbon steels • steel alloys (stainless steel).	Students will not be expected to have practical experience of working with all of these metals/alloys but exam questions could refer to any of the properties listed.	

Ferrous Metals



In this presentation you will:

- investigate types of steel and their applications

Next >

Introduction

Metals that contain ***'iron'*** are known as ***Ferrous Metals***.

Iron and ***steel*** have been used for thousands of years.

Iron and ***steel*** were used to make tools and weapons.

Today, they are still the most important engineering metals.



Next >

Iron



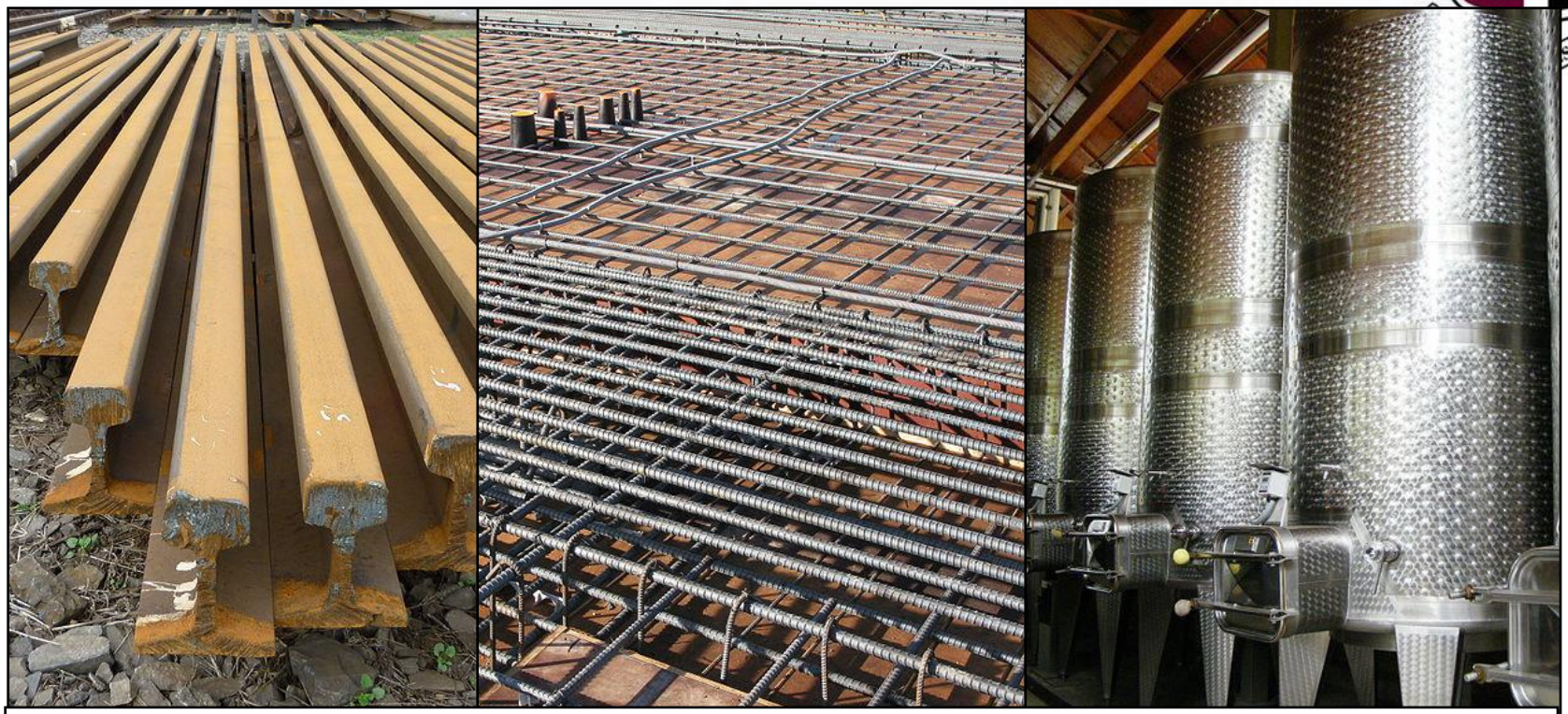
Iron Ore

Iron is an abundant element, usually found as ***iron oxide-based ores*** that must be processed to extract the ***iron***.

Pure iron is a ***soft, ductile*** metal, so it is rarely used in this form.

Next >

Steel

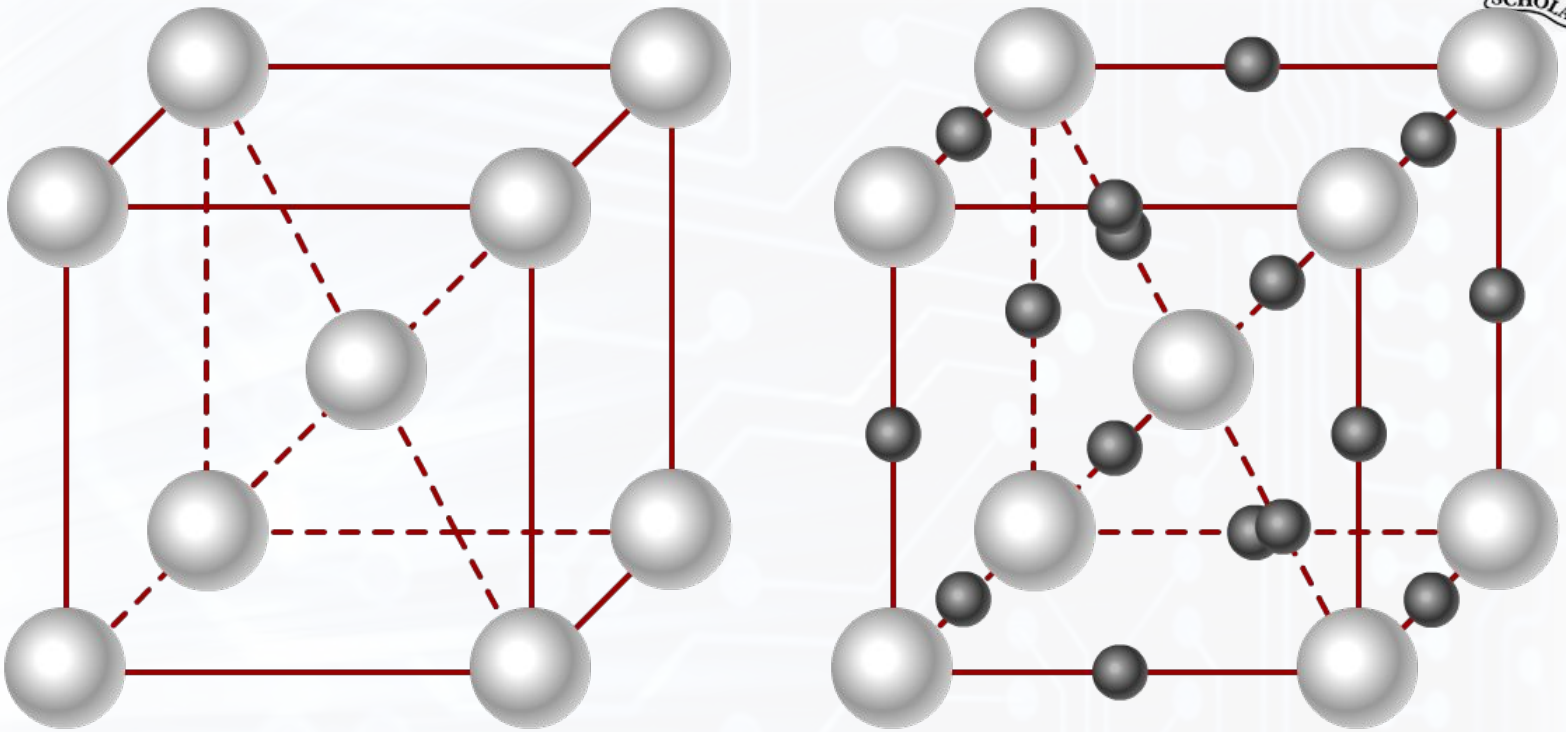


Iron + Carbon = Steel

Iron is used in the form of ***steel***. ***Steel*** is basically an ***alloy*** of ***iron*** and the ***non-metallic element carbon*****.** Some types of ***steel*** have other elements added, but it is the *addition of carbon that turns iron into steel*.

Next >

Steel and Carbon



The addition of just a ***small percentage of carbon*** to ***iron*** can have a *dramatic effect on the resulting alloy*. The *carbon modifies the crystal structure and strengthens it considerably*.

Next >

Low Carbon Steel



Low carbon steel (Mild Steel) has many uses, such as *car bodies, ship hulls and cans*.

It contains up to ***0.15 - 0.35% carbon***. It has a ***low strength***, but is ***inexpensive*** and is ***easy to form*** and ***process***.

Next >

High Carbon Steel







High carbon steel is used for *springs, knives and tools such as punches*.

This type of ***steel*** has ***high strength and hardness***, but is ***brittle***. It contains between ***0.8-1.5% carbon***.

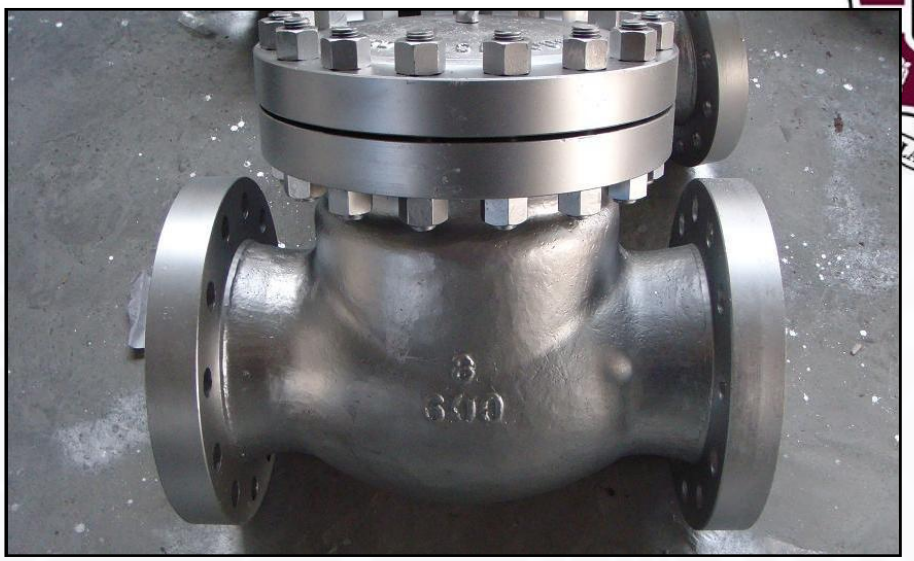
Next >

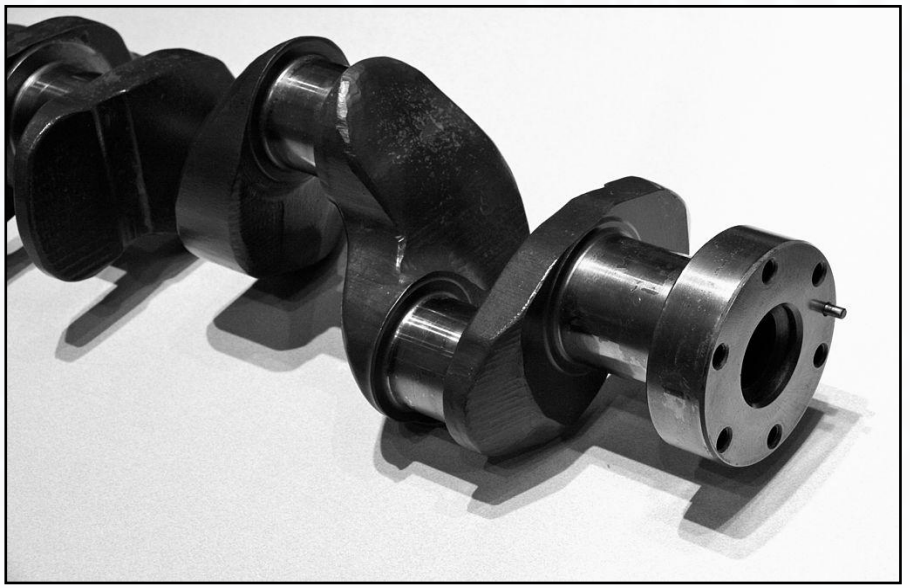
Steel Alloys

In addition to ***iron*** and ***carbon***, *steel alloys contain other elements*.

These added elements give the steel new or improved properties.

Steel alloys are generally stronger, harder, tougher and more durable than ordinary carbon steel.





Next >

Alloying Elements

Alloys can be added to improve properties:

- ***Corrosion resistance – add chromium or nickel***
- ***Machinability – add sulphur, lead or bismuth***
- ***Hardness – add boron, manganese or tungsten***
- ***Strength – add silicon or vanadium***



Next >

Tool Steel







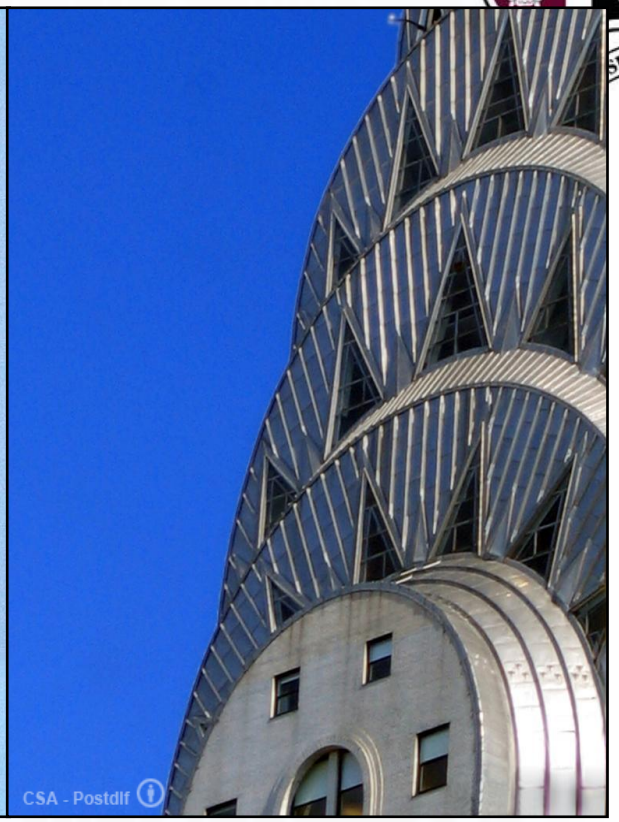
Tool steel - especially ***hard steel alloys*** are used to make *tools, dies and machine parts*.

They include elements such as *nickel, molybdenum or tungsten* to give ***extra hardness*** and ***resistance to wear***.

Next >

Stainless Steel





Stainless steel has a minimum of 11% chromium. The chromium forms a protective layer on the surface of the steel, giving it excellent corrosion-resistant properties.

Next >

Structural Steel





© Steve F

Structural steel is used in *construction*. It is often supplied as *beams with specific cross-sections, such as the 'I' beam*. *Bridges use steel cables*.

Next >

Cast Iron





Cast iron has a ***carbon*** content of ***between 2% and 4%***. Large crystals of carbon, embedded in the matrix of iron, prevent slip. This makes ***cast iron extremely hard and brittle***.

Next >

Cast Steel



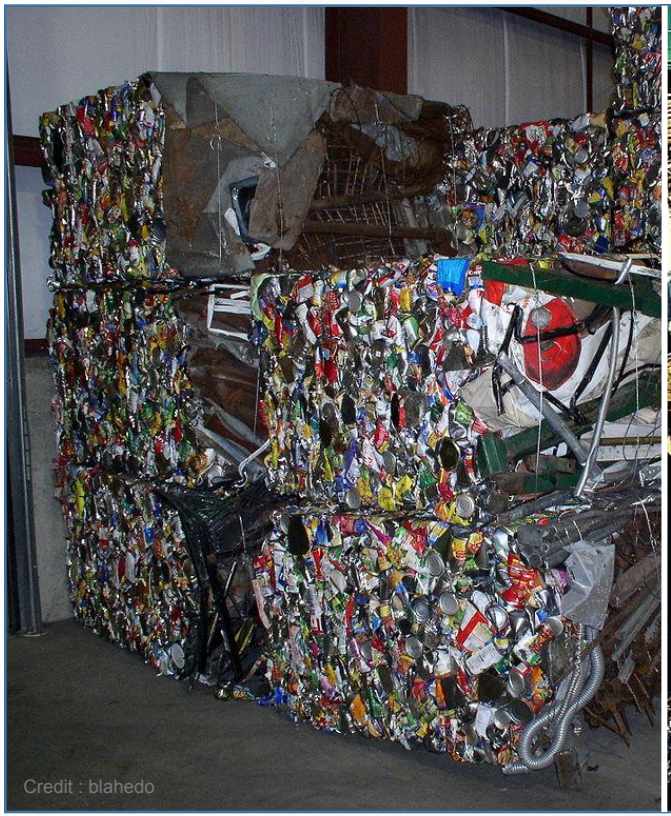


When ***cast iron <u>cannot</u>*** deliver *enough strength or shock resistance, cast steel can be used*.

Cast steel is used to produce *heavy engineering equipment, such as engine casings, mining equipment and turbine wheels*.

Next >

Recycling of Steel





It is more cost effective to recycle steel than to mine iron ore and process it into new steel. Steel does not lose its properties during the recycling process.

Next >

Ferrous Metals – *Properties and Uses*

Name and melting point	Composition	Properties and working characteristics	Uses
Cast iron 1200°C	Iron + 3.5% carbon, wide range of alloys, white, grey and malleable forms	Hard skin, brittle soft core, strong under compression, self lubrication, cannot be bent or forged	Heavy crushing machinery Car brake drums or discs Vices or machine parts
Steel Mild steel 1600°C	Alloys of iron and carbon 0.15–0.35% carbon	Tough, ductile and malleable, high tensile strength, easily joined, welded, poor resistance to corrosion, cannot be hardened and tempered, general purpose material	Nails, screws, nuts and bolts Girders Car bodies
Medium carbon steel	0.4–0.7% carbon	Strong and hard, but less ductile, tough or malleable	Garden tools (trowel, fork) Springs Rails
High-carbon steel (silversteel) 1800°C	0.8–1.5%	Very hard, but less ductile, tough or malleable, difficult to cut, easily joined by carbon heat treatment, strength decreases above 0.9%	Hand tools (hammers chisels, screwdrivers, punches)
Alloy steels Stainless steel	Alloys 18% chromium 8% nickel 8% magnesium	Hard and tough, resists wear, corrosion-resistant, different forms affect malleability (types 18/8), difficult to cut or file	Sinks Cutlery Dishes, teapots
High-speed steel	Medium carbon steel + tungsten + chromium + vanadium	Very hard, resistant to frictional heat even at red heat, it can only be ground	Lathe cutting tools Drills Milling cutters
High tensile steel	Low carbon steel + nickel	Corrosion-resistant, low rate of expansion, exceptional strength and toughness	Gears/engine valves Turbines blades
Manganese steel	1.5% manganese	Extreme toughness	Chains Hooks and couplings

Next >

Question 1

Steel is an alloy of which two elements?

A) Iron and carbon

B) Copper and carbon

C) Iron and copper

D) Iron and bronze

Next >

Question 1

Steel is an alloy of which two elements?



A) Iron and carbon

B) Copper and carbon

C) Iron and copper

D) Iron and bronze

Steel is an alloy of iron and the non-metallic element carbon.

Next >

Question 2

Which of the following has the highest carbon content?

A) Iron ore

B) Low carbon steel

C) High carbon steel

D) Cast iron

Next >

Question 2

Which of the following has the highest carbon content?

A) Iron ore

B) Low carbon steel

C) High carbon steel



D) Cast iron

Cast iron contains between 2% and 4% carbon; all the others listed have a lower carbon content.

Next >

Question 3

Which element is added to stainless steel to improve its corrosion resistance?

A) Carbon

B) Bismuth

✓ C) Chromium

D) Sulphur

Stainless steel has a minimum of 11% chromium to improve corrosion resistance.

Next >

Question 3

Which element is added to stainless steel to improve its corrosion resistance?

A) Carbon

B) Bismuth

✓ C) Chromium

D) Sulphur

Stainless steel has a minimum of 11% chromium to improve corrosion resistance.

Next >

Question 4

Which properties of steel can be improved by adding alloying elements?

A) Corrosion resistance

B) Strength

C) Hardness

D) All of the above

Next >

Question 4

Which properties of steel can be improved by adding alloying elements?

A) Corrosion resistance

B) Strength

C) Hardness



D) All of the above

Next >

Summary

In this presentation you have seen:

- the basic composition of steel
- different types of steel and their applications

End